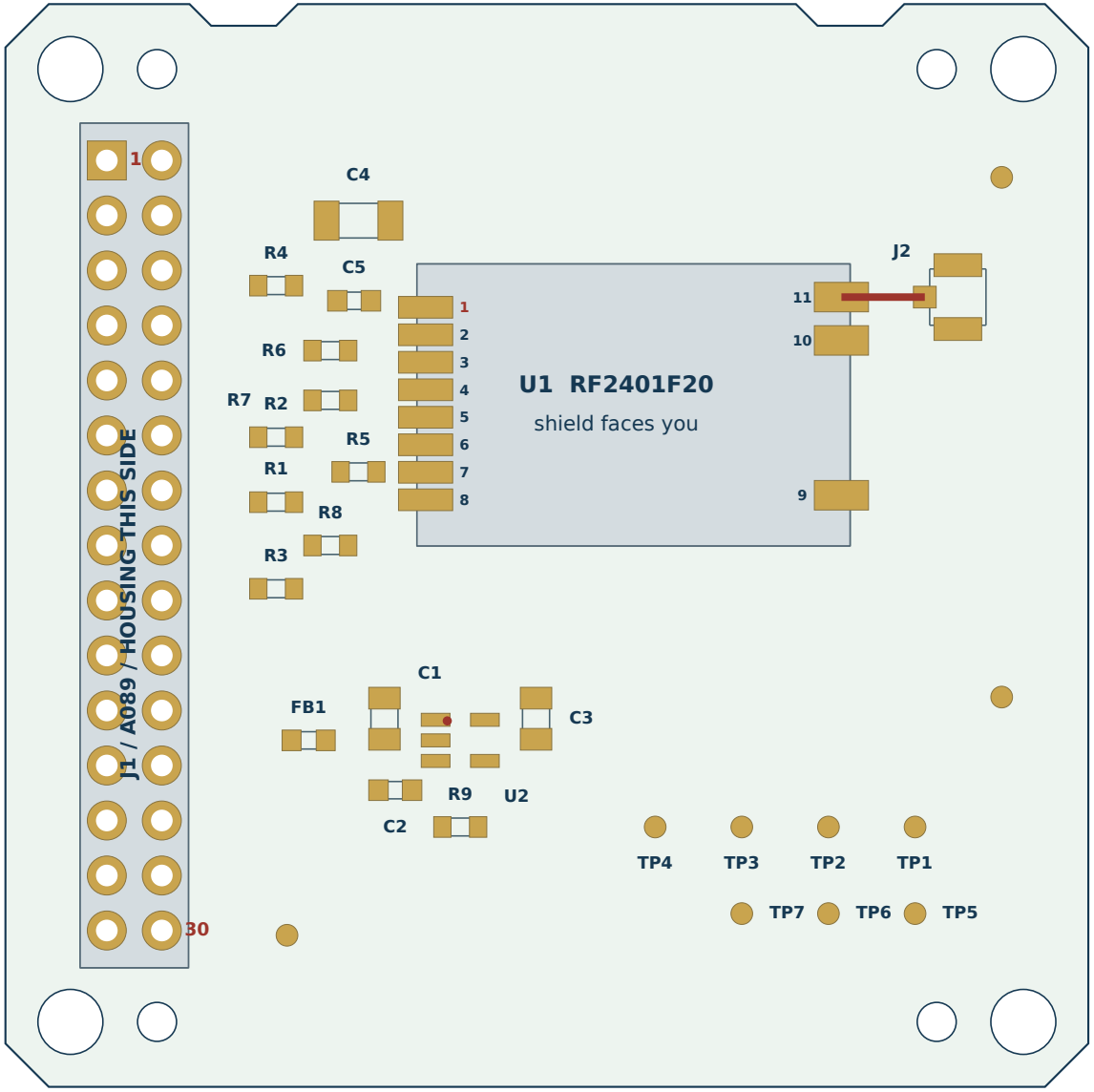
Bottom assembly

View into the expansion housing. Mirrored from normal PCB top coordinates. Drawing scale 3:1.



Populate 18 SMT parts

All radio, power and passive parts mount on B.Cu, facing down into the M032 expansion ring. Use bom.csv and the native footprints for exact parts and land patterns.

Install J1 separately

A089 is a through-hole pass-through socket. Female plastic housing faces you in this drawing. Male tails project through the PCB toward CoreS3 SE. Pin 1 is the square ground pad at the north end.

RF connection

J2 is Hirose U.FL-R-SMT-1(10). U1 pad 11 connects by a straight 3.85 mm trace. Attach Taoglas CAB.628, then GW.11.A153 on the RP-SMA female bulkhead. Add cable strain relief.

Placement convention

positions.csv is native KiCad bottom placement in mm: unmirrored X, negative Y from the top-left board origin, and KiCad bottom rotations. The assembler must map rotations to its machine library.

Inspect before power

Confirm U1 pad 1, U2 pin 1 (red dot), RF pad 11 and all three radio ground pads. No component substitutions for U1. Do not populate the no-connect pad U2.4 with a bridge.

Fit check

Radio body height 2.1 +/-0.1 mm; A089 housing 9.5 mm nominal. Verify PCB seat height, pin engagement, cable bend and all component clearances in the actual housing.

ENGINEERING FIRST ARTICLE - physical fit and native-radio operation not yet validated